Electrochemistry Communications 124 (2021) 106936

Contents lists available at ScienceDirect

Electrochemistry Communications

journal homepage: www.elsevier.com/locate/elecom

A boron-based electrolyte additive for calcium electrodeposition

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ARTICLE INFO

Keywords:
Calcium batteries
Boron-containing additive
Calcium plating
Metal anode

ABSTRACT

In recent years, several organic electrolytes have been reported allowing calcium metal electrodeposition. In all cases, some degree of passivation on the surface of the calcium electrode was reported. Following a recent report on borate-based passivation layer playing a crucial role for Ca plating in electrolyte solutions containing $Ca(BF_4)_2$, here we report the use of a boron-containing additive. The presence of this additive allows Ca plating and stripping using a $Ca(TFSI)_2$ containing electrolyte, which otherwise lead to the formation of a fully Ca^{2+} blocking passivation layer.

1. Introduction

Secondary batteries employing multivalent metal anodes (such as Mg, Ca or Al) are recognized as promising next-generation battery technologies given their high theoretical energy densities [1]. Particularly Ca metal, having a standard redox potential of −2.87 *vs.* SHE and a theoretical specific capacity of 1.34 $Ah \cdot g^{-1}$, would allow the production of high energy density cells competing with the state-of-the-art lithium-ion batteries (LIBs) [2]. However, only few electrolyte solutions have been reported allowing reversible calcium plating/stripping [3–6].

As in the case of LIBs, the low potential of the anode in calcium-metal batteries promotes the reductive decomposition of the components of the electrolyte solution – solvent(s), salt(s) and/or additive(s). The decomposition products will accumulate on the surface of the electrode and ideally form a solid-electrolyte interphase (SEI) which allows Ca^{2+} migration and prevents further electrolyte decomposition.

Recently, we have identified a borate-based SEI component, formed using an organic electrolyte containing $Ca(BF_4)_2$ salt, as a promising candidate for stable Ca-metal plating [7]. While the exact structure/stoichiometry of these borate species has not yet been fully elucidated, it is clear that they are the result of anion decomposition. As an alternative to the use of $Ca(BF_4)_2$, which is commercially available only as a hydrate and causes problems during drying [8], we explore the use of a boron containing additive into an easily dried calcium bis(trifluoromethyl) sulfonimide ($Ca(TFSI)_2$) electrolyte.

Trifluoroborate complexes ($BF_3 \cdot X$, X = cyclic or linear carbonate) were employed recently as additives for LIBs in $Li[Ni_xCo_yMn_zO_2]||$ graphite cells [9]. Higher capacity retention and lower cell impedance were reported when the additives were used with the $LiPF_6$ (1 M) in EC: EMC (3:7 wt%) electrolyte. While the SEI formed on the graphite electrode was not fully characterized, elemental analysis after cycling evidenced up to 8 at.% of boron, suggesting the presence of the trifluoroborate complex decomposition products in the SEI.

Here, we investigate the use of a commercial trifluoroborate complex as additive in calcium-metal battery electrolytes. The reductive stability of electrolytes based in $Ca(TFSI)_2$, containing different concentrations of trifluoroborate-diethyletherate ($BF_3 \cdot DE$), is reported. The influence of the additive content and of cycling conditions are also evaluated to find the optimal operation parameters for the Ca plating/stripping.

2. Experimental

All electrochemical tests were performed at 100 °C using Swagelok© T-type cells with a stainless steel disk as working electrode, silver wire as pseudo-reference (Ag_{QRE}) and a calcium disk as counter electrode. The stainless steel disks (SS316, Ø10 mm, Goodfellow) were rinsed with ethanol in an ultrasonic bath and dried at 70 °C during 12 h before use. Silver wire (AlfaAesar) was polished and rinsed in nitric acid 1 $mol \cdot L^{-1}$ for 3 min. Calcium disks were obtained by pressing redistilled calcium granules (99.5%, AlfaAesar) and scratching the surface to remove any passivating surface layer. Whatman glass fiber disks were used as separators.

Boron trifluoride diethyletherate ($BF_3 \cdot DE$, >46.5% BF_3, Sigma Aldrich) was used as additive without further purification. *Note that BF_3 is a Lewis acid neutralized with the DE. It is a corrosive and irritating liquid. The reaction with water is violent and can generates BF_3 gas and hydrofluoric*

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https://doi.org/10.1016/j.elecom.2021.106936
Received 30 December 2020; Received in revised form 14 January 2021; Accepted 17 January 2021
Available online 21 January 2021

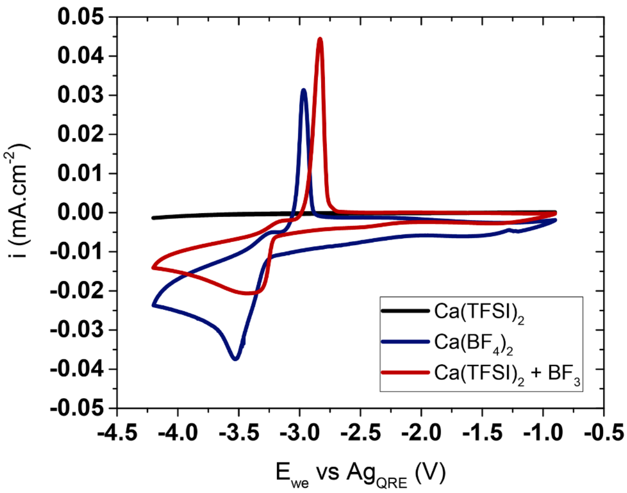

Fig. 1. Cyclic voltammetry of a Ca(TFSI)$_2$ 0.4 mol·L^{-1} EC:PC electrolyte (black line), with the addition of 2 wt% of BF_3·DE additive (red line), and with Ca $(BF_4)_2$ 0.4 mol·L^{-1} EC:PC electrolyte (blue line) (100 °C, 0.1 mV·s^{-1}, cycle number 6). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

acid, a highly corrosive acid which causes deep burns. The BF_3 is a highly toxic gas (exposure limit value is set at 1 ppm).

Different concentrations of the additive were added to an electrolyte solution of 0.4 mol·L^{-1} Ca(TFSI)$_2$ (calcium bis(trifluoromethanesulfonyl)imide, 99,9% Solvionic) in a 50/50 wt% mixture of propylene carbonate (PC, from Aldrich, anhydrous, 99.7%) and ethylene carbonate (EC, from Aldrich, anhydrous, 99.0%). The solvent mixture was previously dried over molecular sieves (~3 Å, Alfa Aesar) for at least two days before the preparation of the electrolytes. The water content of the electrolyte before the addition of BF_3·DE was below 5 ppm (measured by Karl Fischer titrations inside a glove box). As the additive reacts with traces of water, the final electrolyte solutions used in this study are not expected to contain higher water content. A total of 450 µL of electrolyte was used in each cell. All cells were assembled inside an argon filled glovebox with O_2 and H_2O concentrations below 1 ppm and tested using a Bio-Logic VSP300 potentiostat.

Scanning electron microscopy (SEM) studies were performed using a Quanta 200 ESEM FEG FEI microscope. Samples were transferred from an argon filled glove box with minimum air exposure.

3. Results and discussion

The cyclic voltammograms (CV) recorded using 0.4 mol·L^{-1} Ca (TFSI)$_2$ in EC:PC without (black trace) or with 2 wt% of BF_3·DE (red trace) are plotted in Fig. 1. For comparison, an electrolyte of 0.4 mol·L^{-1} Ca(BF$_4$)$_2$ in the same solvent mixture is included (blue trace). As reported earlier [3,7] the use of Ca(TFSI)$_2$ in EC:PC results in the absence of any significant redox features. This behavior was ascribed to the Ca^{2+} blocking nature of the passivation layer being formed in this electrolyte [7]. By contrast, using an electrolyte containing Ca(BF$_4$)$_2$ a clear reduction wave can be observed with an onset potential at about −3.25 V *vs.* Ag_{QRE}. This reduction process, ascribed to Ca plating, is at least partially reversible and the associated oxidation peak (Ca stripping) has an onset potential at *ca.* −3.12 V *vs.* Ag_{QRE}. The addition of BF_3·DE to the Ca(TFSI)$_2$-based electrolyte (red trace) resulted in similar redox features than for the Ca(BF$_4$)$_2$ containing electrolyte and Ca plating and stripping is thus taking place when this additive is used. The characterization of the SEI being formed using BF_3·DE is out of the scope of the present work and will be systematically investigated in a forthcoming study. However, the fact that Ca plating is feasible in Ca(TFSI)$_2$ + BF_3·DE suggests that this additive contributes to the formation of a Ca^{2+} permeable SEI similar to the one obtained with BF_4^-. Tests were also performed at 30 °C but no significant plating stripping could be recorded, similar to what is happening with the Ca(BF$_4$)$_2$ EC:PC electrolyte [3].

The morphology of the electrodes resulting from a potentiostatic deposition (−4.2 V, 72 h, 100 °C) with or without additive was investigated by SEM (Fig. 2). A smooth and conformal deposit could be observed at the surface of the stainless steel substrate used as the working electrode when no additive was used. While the thickness of such deposit cannot be accurately estimated, it is expected to be extremely thin as the general morphology of the substrate is maintained. This is in agreement with our previous work [7] in which using transmission electron microscopy, deposit thickness below 20 nm was estimated using this electrolyte. When BF_3·DE is used, significantly thicker deposit was observed, presenting several cracks and did not uniformly cover the surface of the stainless steel disk.

Few differences can be observed between the cyclic voltammograms obtained with Ca(BF$_4$)$_2$ and Ca(TFSI)$_2$ + BF_3·DE electrolytes. For instance, Ca plating and stripping processes are centered at slightly different potentials depending on the electrolyte being used (−3.10 and −3.19 V *vs.* Ag_{QRE}, respectively in Ca(TFSI)$_2$ + BF_3·DE and Ca(BF$_4$)$_2$ electrolytes). However, this potential shift is most likely related to the poor reliability of the silver reference electrode in organic electrolytes [10,11]. By contrast with the Ca(TFSI)$_2$ + BF_3·DE system, a significant residual reduction current shifting the entire CV curve below the zero

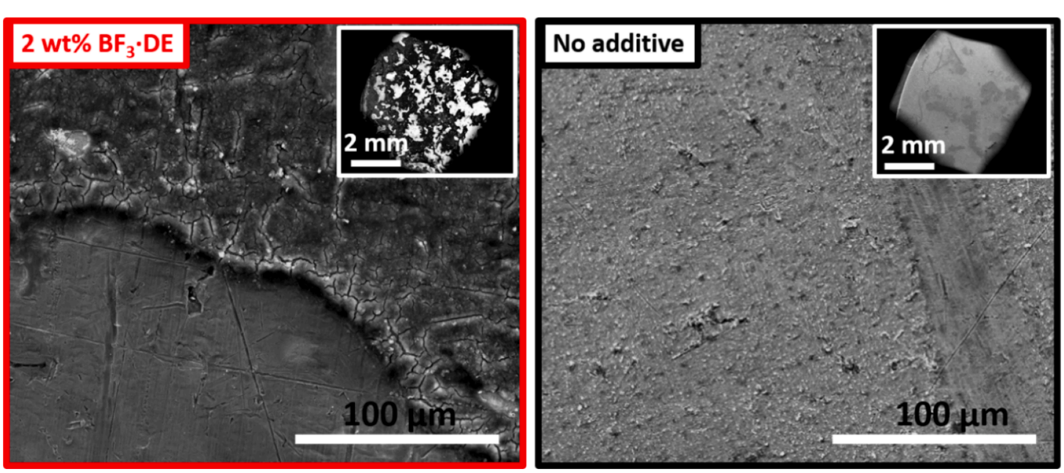

Fig. 2. Scanning electron microscopy images of stainless steel substrates after 72 h deposition at −4.2 V *vs.* Ag_{QRE} (100 °C) using Ca(TFSI)$_2$ 0.4 mol·L^{-1} EC:PC electrolyte with or without 2 wt% of BF_3·DE additive.

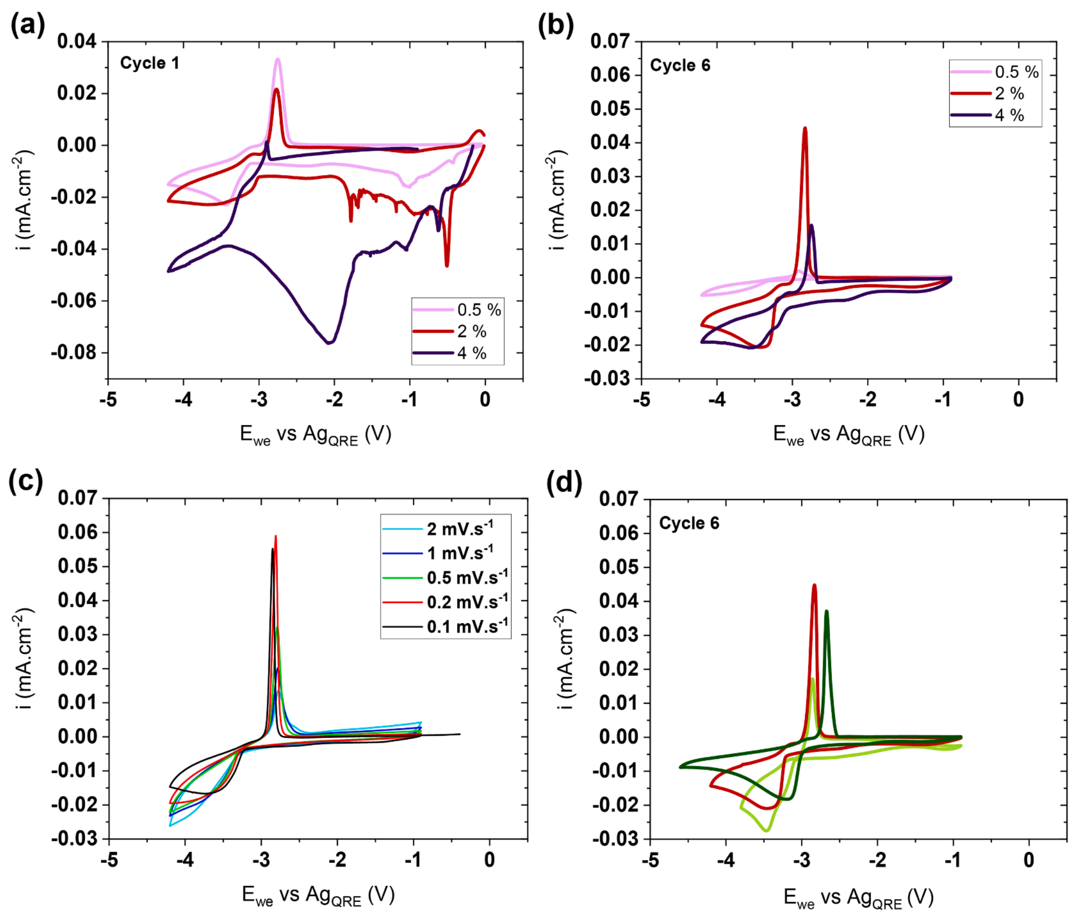

Fig. 3. Cyclic voltammetry at 100 °C of a $Ca(TFSI)_2$ 0.4 $mol{\cdot}L^{-1}$ EC:PC electrolyte with the addition of 2 wt% of diethyletherate (black line) or $BF_3{\cdot}DE$ at 0.5 wt% (pink line), 2 wt% (red line) and 4 wt% (purple line) 0.1 $mV{\cdot}s^{-1}$ for (a) the first cycle and (b) the sixth cycle. (c) Cycles with different scan rates (fifth cycle) with 2 wt % $BF_3{\cdot}DE$ in the electrolyte and (d) with different potential windows (0.1 $mV{\cdot}s^{-1}$, sixth cycle). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

current line is observed when $Ca(BF_4)_2$ based electrolyte is used (Fig. 1). Indeed, with $Ca(TFSI)_2$ + $BF_3{\cdot}DE$, lower (if any) residual reduction current was observed, mostly during the first few cycles and will be discuss in more details below.

In order to better understand the impact of $BF_3{\cdot}DE$ on Ca plating and stripping, several weight contents of the additive were tested. CVs obtained with 0.5 (pink trace), 2 (red trace) or 4 wt% (purple trace) of $BF_3{\cdot}DE$ are shown in Fig. 3a and b. $BF_3{\cdot}DE$ contents of 0.1 and 8 wt% were also evaluated but no plating/stripping could be achieved (data not shown). Upon the first cathodic scan and for all $BF_3{\cdot}DE$ contents, a sharp reduction peak is observed at about −0.5 V *vs.* Ag_{QRE} followed by broader reduction waves at lower potentials (Fig. 3a). The overall charge (in mC) corresponding to these is mostly irreversible. The reduction processes increase with the amount of additive in solution with about −340, −400, −700, −1680 and −3320 mC for 0.1, 0.5, 2, 4, and 8 wt% of $BF_3{\cdot}DE$ in solution respectively. Upon the first anodic scan, clear Ca stripping peaks were only achieved with 0.5 and 2 wt% $BF_3{\cdot}DE$ solutions. For electrolytes containing 2 wt% or more, a residual reduction current can be seen after the Ca stripping peak between −2.7 and −0.9 V *vs.* Ag_{QRE}.

Upon cycling, this residual reduction current significantly decreased, indicating the formation of an effective passivation layer and/or the consumption of the additive in solution. In the meantime, the Ca stripping peaks evolved differently depending on the initial additive content. The stripping current decreased significantly after six cycles in the solution containing 0.5 wt% $BF_3{\cdot}DE$. Further cycling resulted in featureless CV curves pointing at the formation of fully blocking passivation layers, similar to the 0.1 wt% $BF_3{\cdot}DE$ or the pure $Ca(TFSI)_2$ electrolytes. For higher additive content (2 wt% and above) an increase in the stripping capacity was recorded upon the first few cycles. Overall, the electrolyte containing 2 wt% of $BF_3{\cdot}DE$ was found to present the highest and most stable calcium stripping charge upon cycling and was selected for further tests.

The influence of the sweep rate (Fig. 3c) and of the lower cutoff voltage (Fig. 3d) on the calcium electrodeposition was also investigated. After ten cycles performed at 0.1 $mV{\cdot}s^{-1}$, leading to an almost complete disappearance of the residual reduction current between −2.7 and −0.9 V *vs.* Ag_{QRE}, the sweep rate was increased to 0.2, 0.5, 1, and 2 $mV{\cdot}s^{-1}$, the fifth cycles of each rate are shown in Fig. 3c. As the sweep rate was increased, the onset potentials for Ca plating and stripping were found to be shifted to, respectively, lower and higher potential values. The voltage hysteresis between plating and stripping (difference between the onset potentials) were found to increase from 200 mV up to 400 mV when the sweep rate is changed from 0.1 to 2 $mV{\cdot}s^{-1}$. In addition, as the sweep rate increased the stripping peak current intensity was found to decrease and a significant peak broadening was observed, indicating a rather poor rate capability.

Calcium plating and stripping is expected to be significantly affected by any impedance building up during cycling. In order to ascertain how calcium plating overpotentials could affect the cyclability of the metal anode, different lower cutoff voltages were evaluated (−3.8, −4.2 and −4.6 V *vs.* Ag_{QRE}, Fig. 3d). The first effect observed by setting a lower cutoff voltage is a decrease in residual reduction current (between −2.7 and −0.9 V *vs.* Ag_{QRE}), indicating that such irreversible reduction

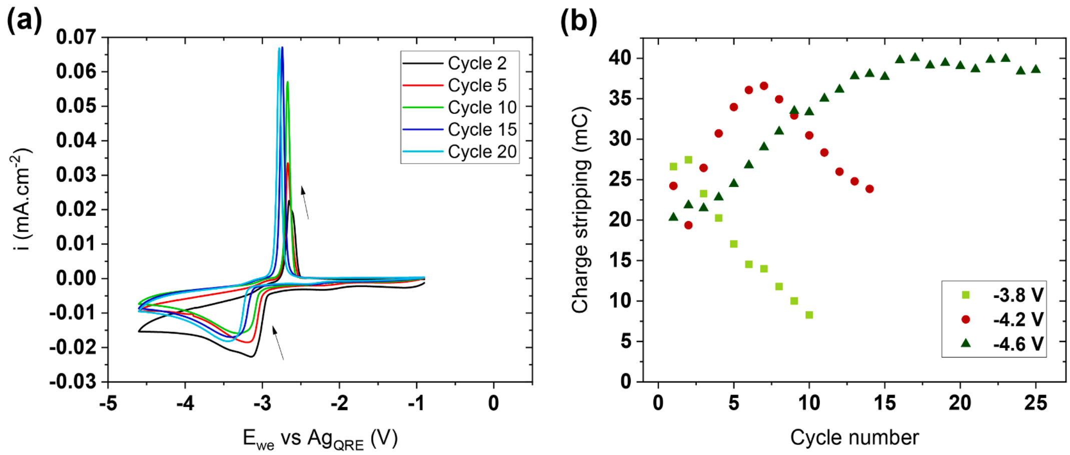

Fig. 4. (a) Cyclic voltammetries at 0.1 mV·s^{-1} of a $Ca(TFSI)_2$ 0.4 mol·L^{-1} EC:PC electrolyte at 100 °C, with 2 wt% of BF_3·DE (b) The coulombic efficiency is exhibiting depending on the number of cycles for each potential window [−0.9 V; −3.8 V *vs.* Ag_{QRE}] (light green squares), [−0.9 V; −4.2 V *vs.* Ag_{QRE}] (red circles), [−0.9 V; −4.6 V *vs.* Ag_{QRE}] (dark green triangles). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

process is taking place concomitantly with Ca plating until a stable passivation layer is formed and/or the additive is fully reduced.

The second effect of the lower cutoff voltage is observed in the evolution of the charge associated with calcium stripping upon cycling. For the first three cycles, rather similar stripping charge were recorded regardless of the cutoff voltage (between 20 and 27 mC, Fig. 4b). However, a significant drop of the stripping charge was observed for the cell cycling down to −3.8 V *vs.* Ag_{QRE}. An increase of the stripping charge was observed for −4.2 V *vs.* Ag_{QRE} cutoff until the seventh cycle followed by a drop whereas increase and stabilization was recorded for the cells cycled down to −4.6 V *vs.* Ag_{QRE}. These results suggest that Ca plating is still taking place well below the potential at which Ca^{2+} mass transport limitation results in the presence of a reduction peak (around −3.6 V *vs.* Ag_{QRE}).

While the origins of the stripping charge decay remain unclear, two possible reasons can be foreseen. First, considering the poor reliability of the silver pseudo reference electrode, any significant change in the electrolyte could results in a potential shift which could affect Ca plating. In some cases, an overall shift of the CV curves was indeed observed towards lower potential values (Fig. 4a). Also, any increased overpotential for Ca plating during cycling would eventually result in limited Ca electrodeposition (depending on the lower cutoff voltage). Nonetheless, for a cutoff voltage of −4.6 V *vs* Ag_{QRE} a continuous increase of the stripping charge was recorded up to cycle number 15 with about 40 mC before it stabilized (Fig. 4b).

In all cases, very low coulombic efficiencies for Ca plating and stripping were obtained (about 30% or below). While the origins behind the high irreversible reduction current is still unclear, the fact that increasing the sweep rate reduced further the coulombic efficiency suggests that the parasitic reaction taking place is kinetically favored over Ca electrodeposition. This is in stark contrast with what is commonly observed in Li cells in which parasitic electrolyte decomposition tends to be mitigated at high C rates [12]. Here, it is expected that both the poor Ca plating/stripping kinetics and the high overpotential required to reach stable cycling, favored concomitant parasitic reduction of the electrolyte.

4. Conclusion

In this study, we report that the addition of 2 wt% of borontrifluoride diethyletherate to a 0.4 mol·L^{-1} $Ca(TFSI)_2$ EC:PC electrolyte allows for calcium plating and stripping. The low coulombic efficiency for calcium plating/stripping (below 30%) obtained in the conditions of this study point at calcium deposition occurring concomitantly with a parasitic reduction reaction, the latter being kinetically favored. Although the exact mechanism enabling the Ca^{2+} diffusion through the passivation layer and the composition of the latter are not fully understood, this report confirms the crucial role of boron containing species in the electrolyte (as the salt or an additive). These results pave the way for further investigations aiming at improving Ca plating/stripping reversibility and to better understand the passivation layer formed in these electrolytes.

CRediT authorship contribution statement

Juan Forero-Saboya: Conceptualization, Methodology, Writing - review & editing. **Charlotte Bodin:** Investigation, Methodology, Writing - review & editing. **Alexandre Ponrouch:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Funding from the European Union's Horizon 2020 research and innovation program H2020 is acknowledged: European Research Council (ERC-2016-STG, CAMBAT grant agreement no. 715087) and H2020-MSCA-COFUND-2016 (DOC-FAM, grant agreement no. 754397). A. Ponrouch is grateful to the Spanish Ministry for Economy, Industry and Competitiveness Severo Ochoa Programme for Centres of Excellence in R&D (CEX2019-000917-S). This work has been done in the frame of the Doctoral Degree Program in Materials Science by the Universitat Autònoma de Barcelona. Alistore-European Research Institute is gratefully acknowledged for financial support through the postdoc grant to C. Bodin.

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